

Task 0 – CNN

Input Image :

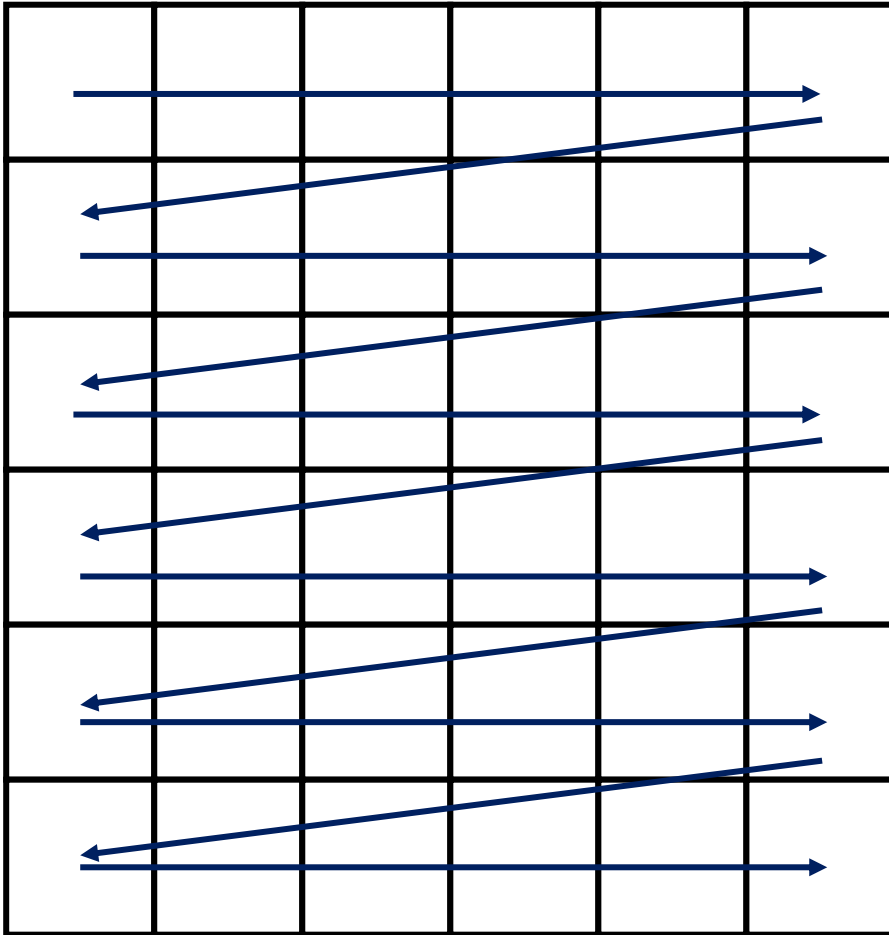


Image 0

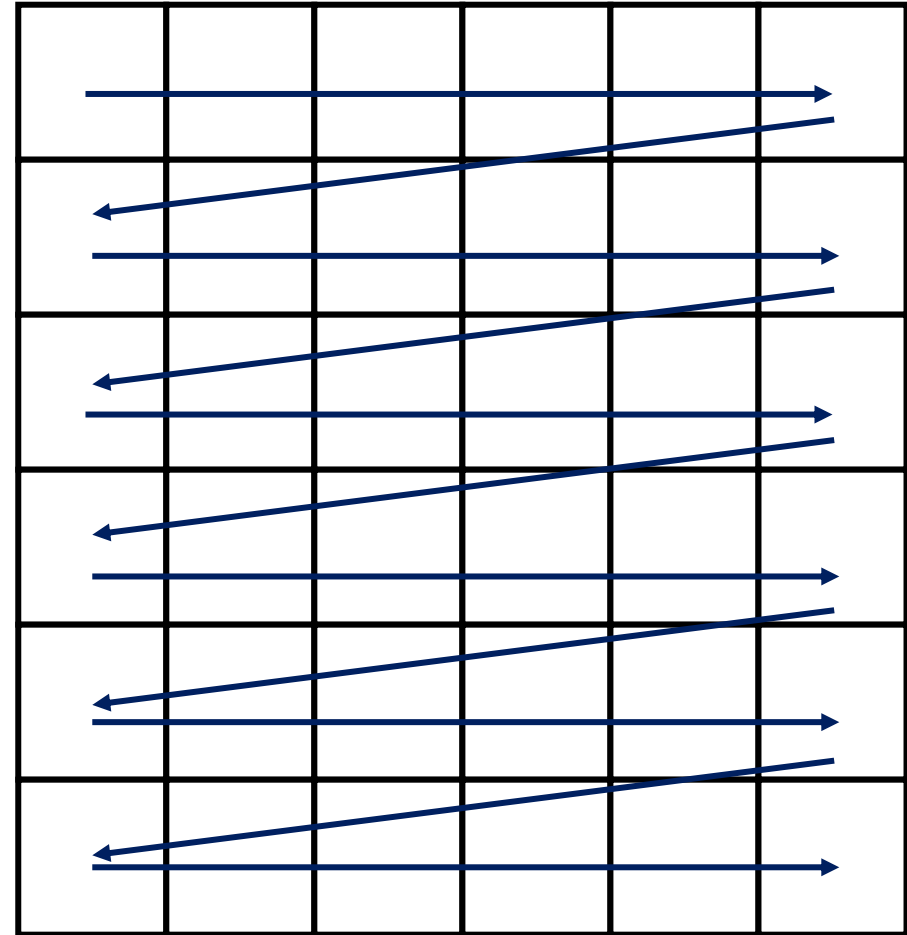
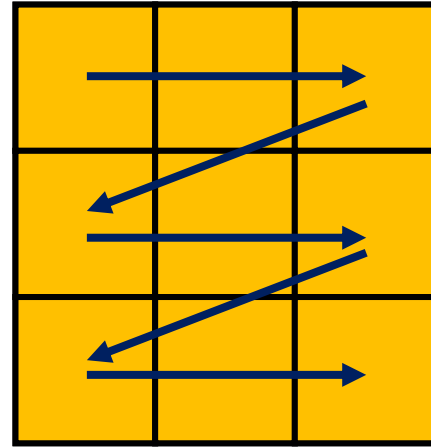
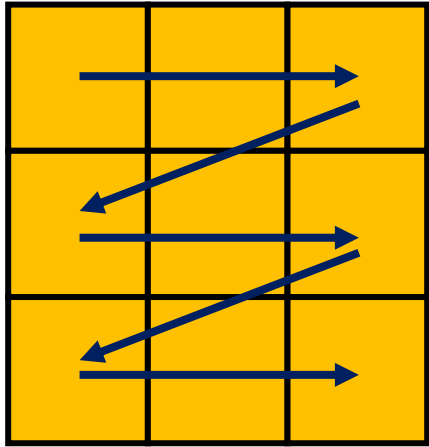


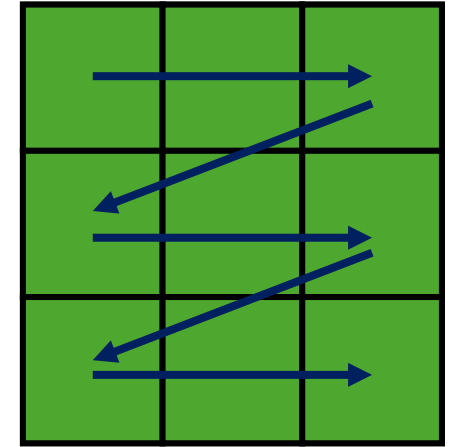
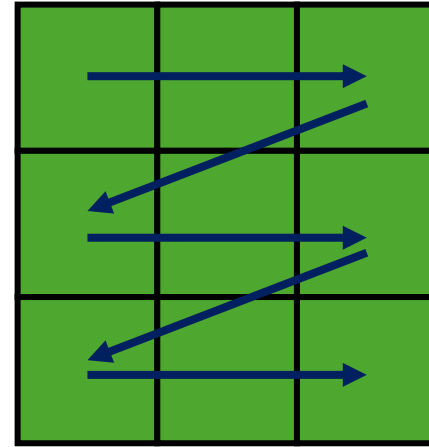
Image 1

Input 2 channel Kernel :

Kernel_ch1



Kernel_ch2



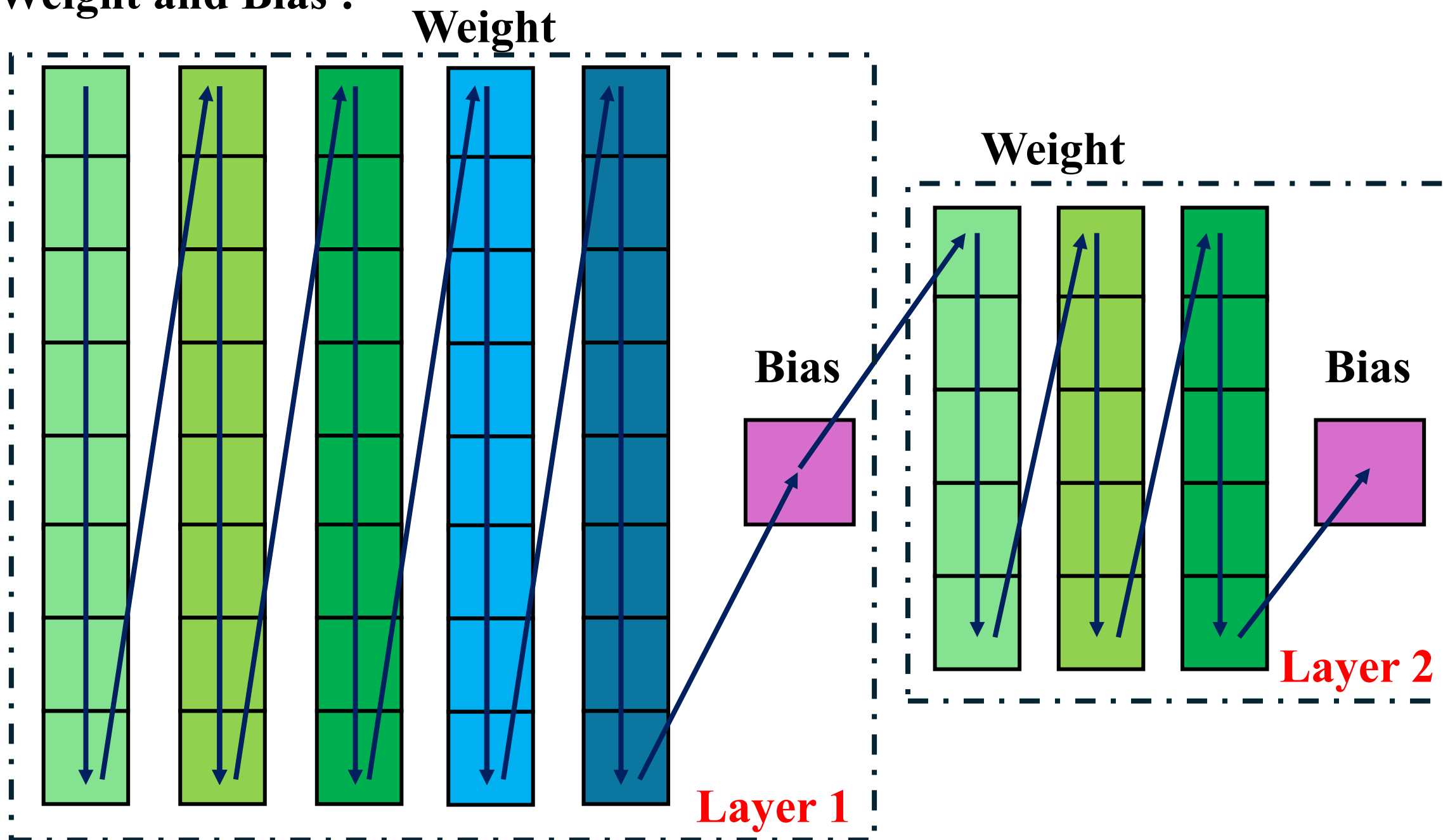
Kernel_Channel 1_1

Kernel_Channel 1_2

Kernel_Channel 2_1

Kernel_Channel 2_2

Input Weight and Bias :



Replication Padding
(mode = 2'b00 or 2'b01) :

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36

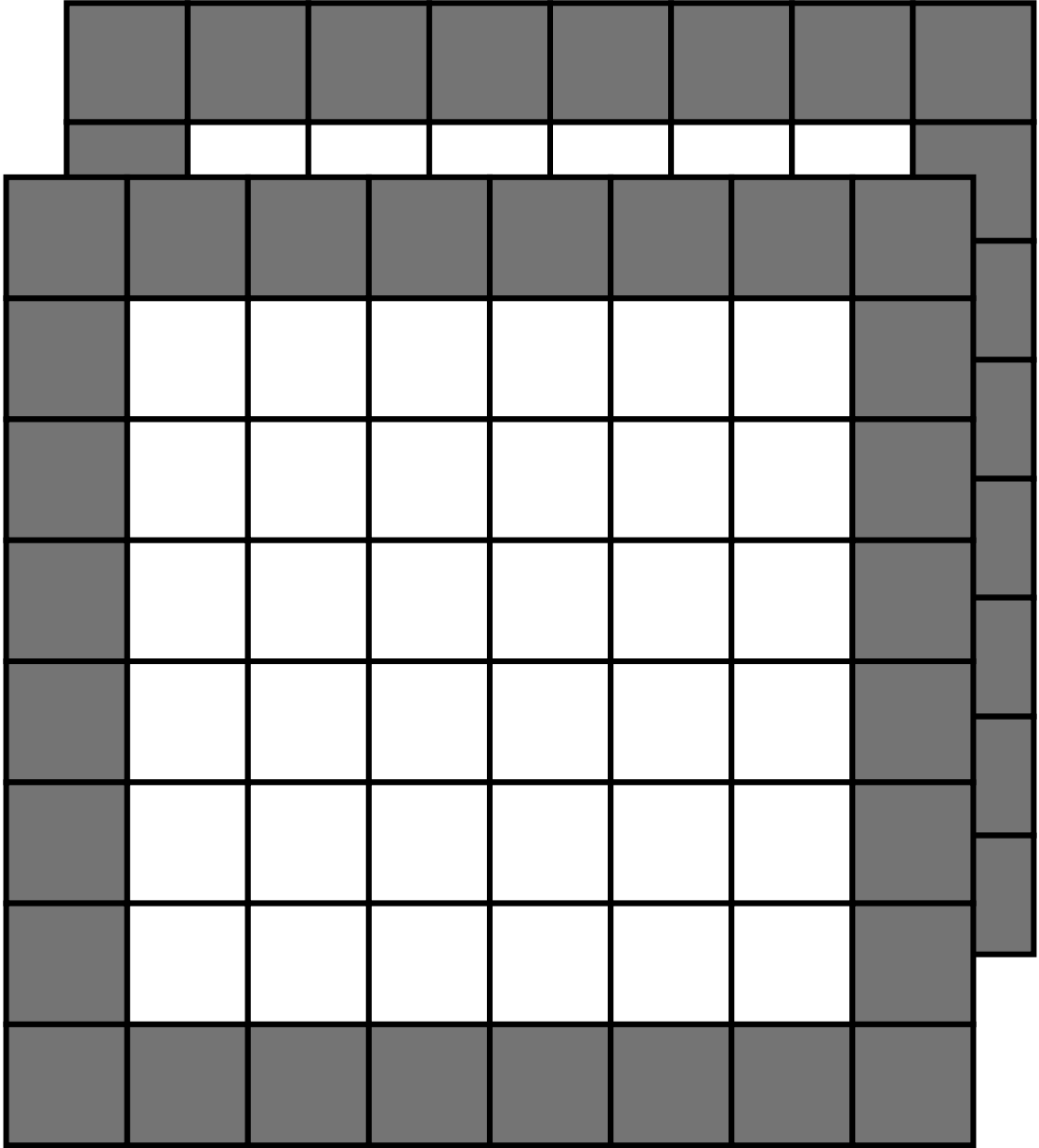
1	1	2	3	4	5	6	6
1	1	2	3	4	5	6	6
7	7	8	9	10	11	12	12
13	13	14	15	16	17	18	18
19	19	20	21	22	23	24	24
25	25	26	27	28	29	30	30
31	31	32	33	34	35	36	36
31	31	32	33	34	35	36	36

Reflection Padding
(mode = 2'b10 or 2'b11) :

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36

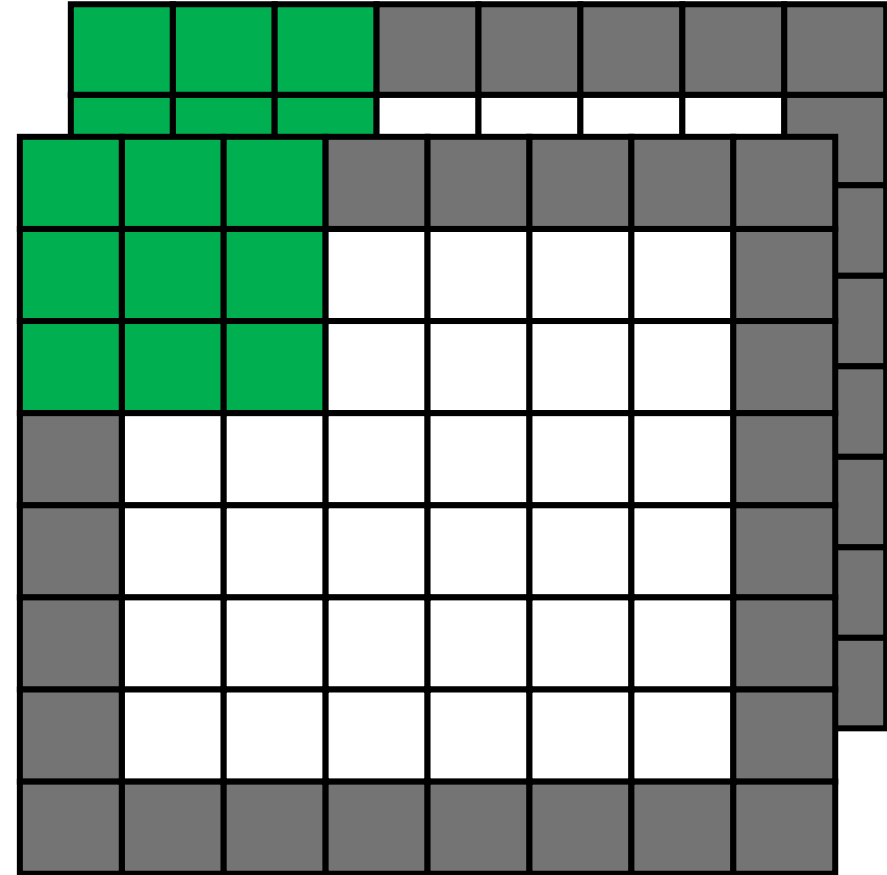
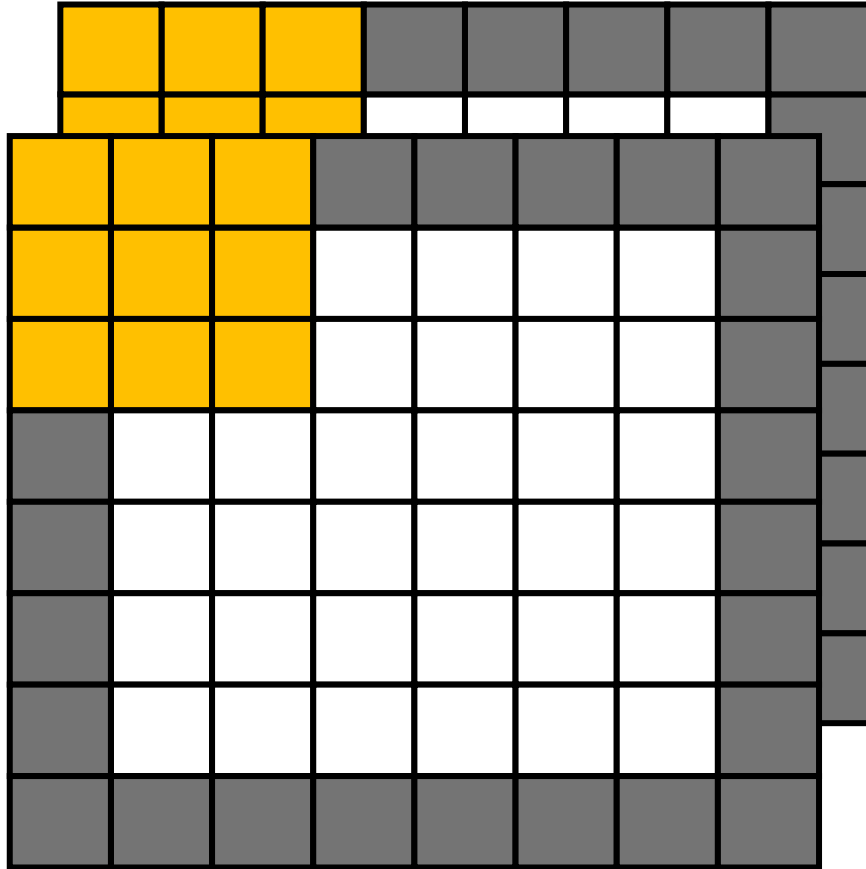
8	7	8	9	10	11	12	11
2	1	2	3	4	5	6	5
8	7	8	9	10	11	12	11
14	13	14	15	16	17	18	17
20	19	20	21	22	23	24	23
26	25	26	27	28	29	30	29
32	31	32	33	34	35	36	35
26	25	26	27	28	29	30	29

After Padding :



(8 x 8, 2 channel)

CNN :



CNN :

1	1	2	3	4	5	6	6
1	1	2	3	4	5	6	6
7	7	8	9	10	11	12	12
13	13	14	15	16	17	18	18
19	19	20	21	22	23	24	24
25	25	26	27	28	29	30	30
31	31	32	33	34	35	36	36
31	31	32	33	34	35	36	36

8 x 8 Image (After Padding)

$1*1 + 1*1 + 2*1 + 1*1 + \dots + 8*1 = 30$

$1*1 + 2*1 + 3*1 + 1*1 + \dots + 9*1 = 36$

x

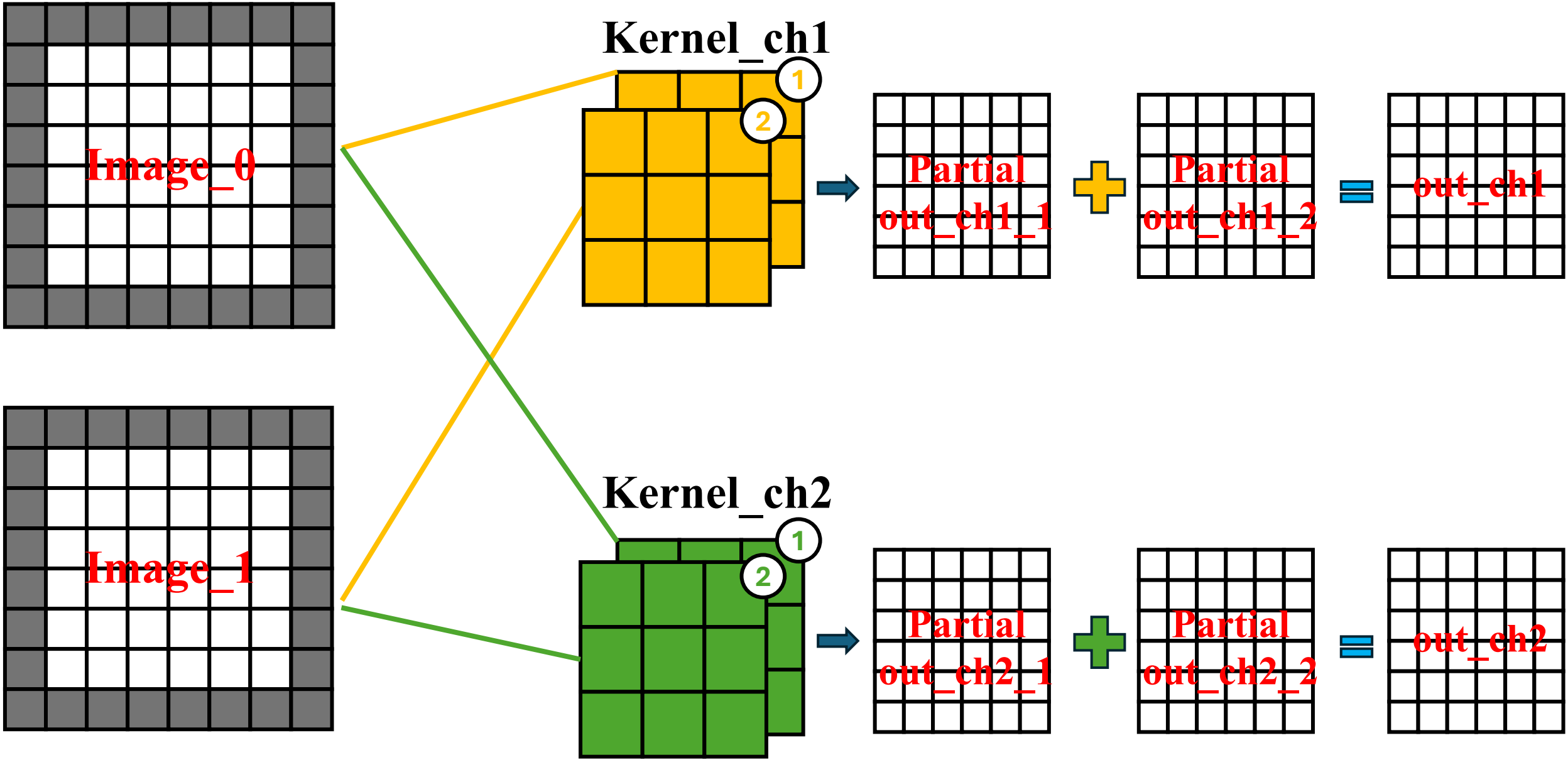
1	1	1
1	1	1
1	1	1

3 x 3 Kernel

=

30	36				

6 x 6 out



Max Pooling :

2	3	5	12	5	7
5	6	3	9	2	2
8	10	1	15	1	4
7	2	13	2	4	9
9	1	6	4	7	1
12	3	2	5	10	13



Max Pooling



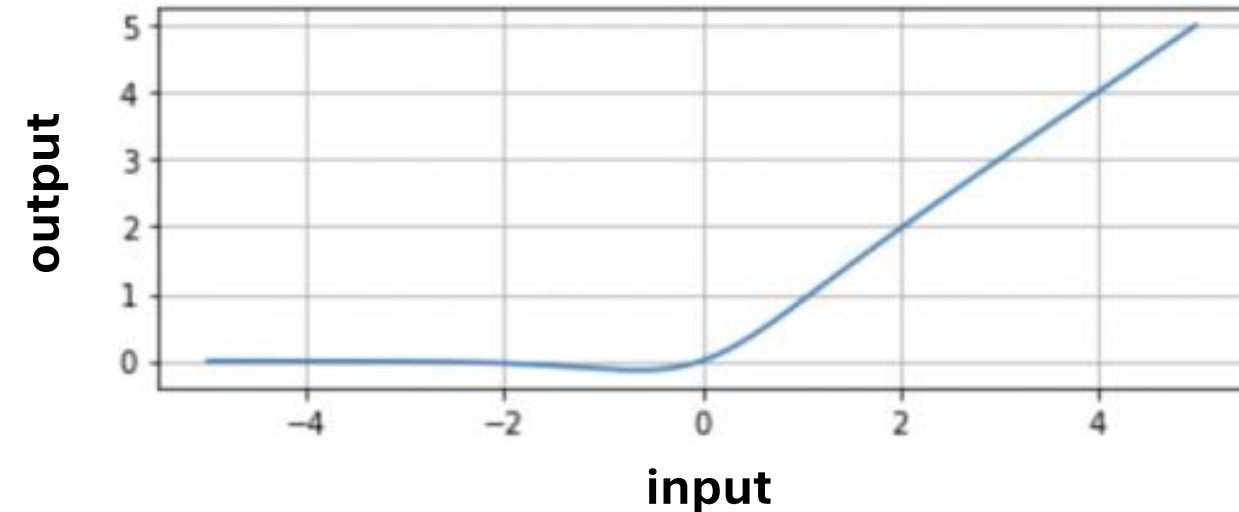
10	15
13	13

Activation function

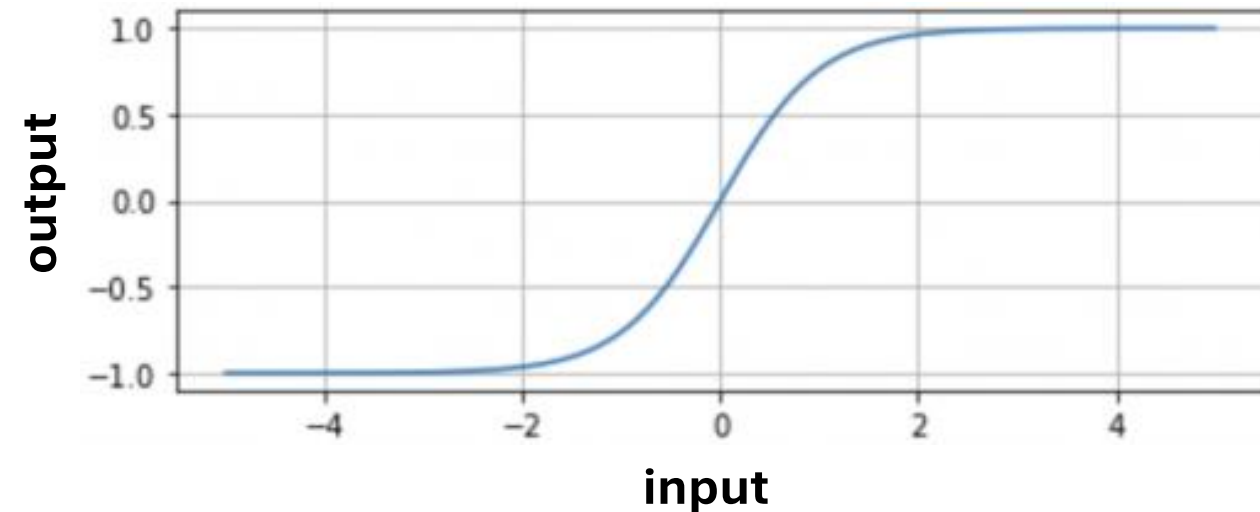
(mode = 2'b00 or 2'b10 for **tanh**, mode = 2'b01 or 2'b11 for **swish**) :

	5	6
1	2	3
3	4	

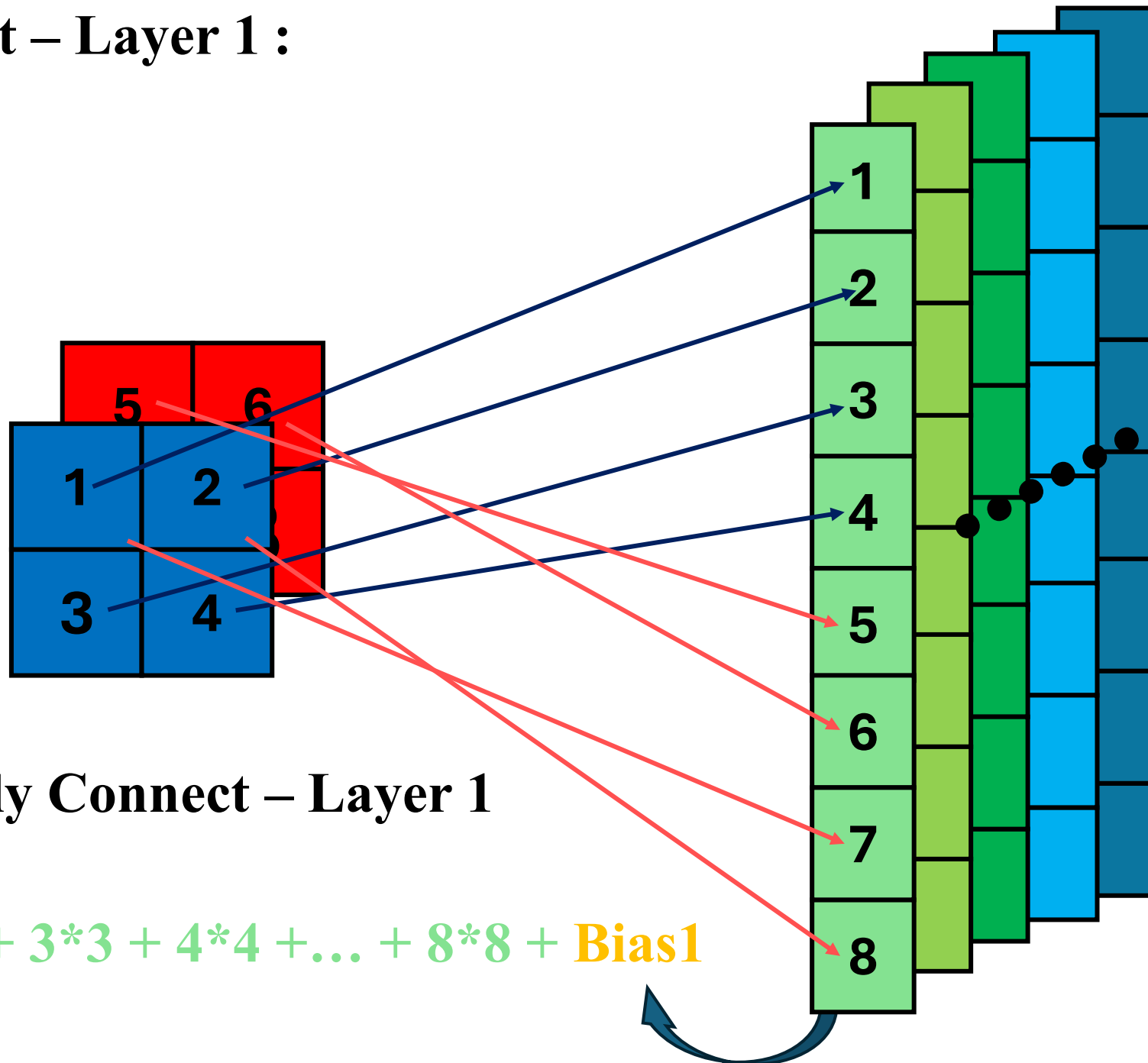
$$\textit{swish}(x) = \frac{x}{1 + e^{-x}}$$



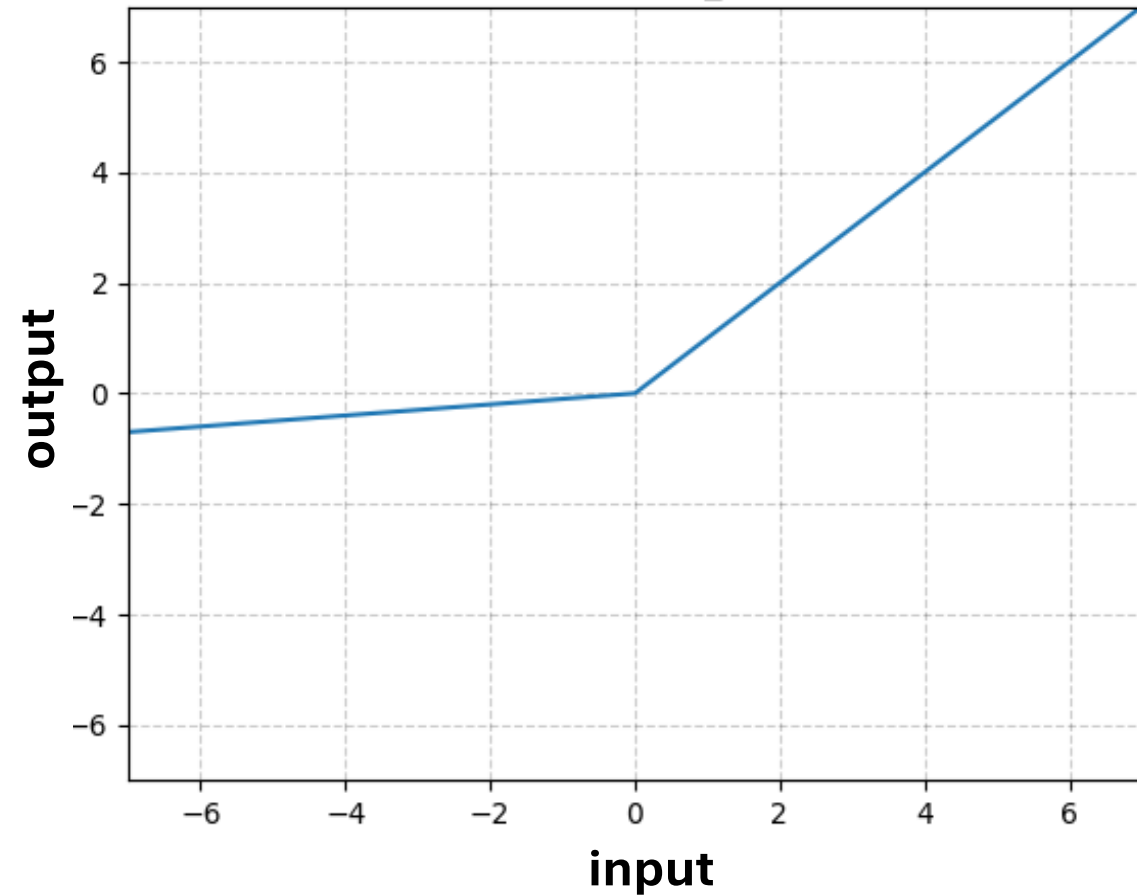
$$\textit{tanh}(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



Fully Connect – Layer 1 :



Activation function (Leaky ReLU) :



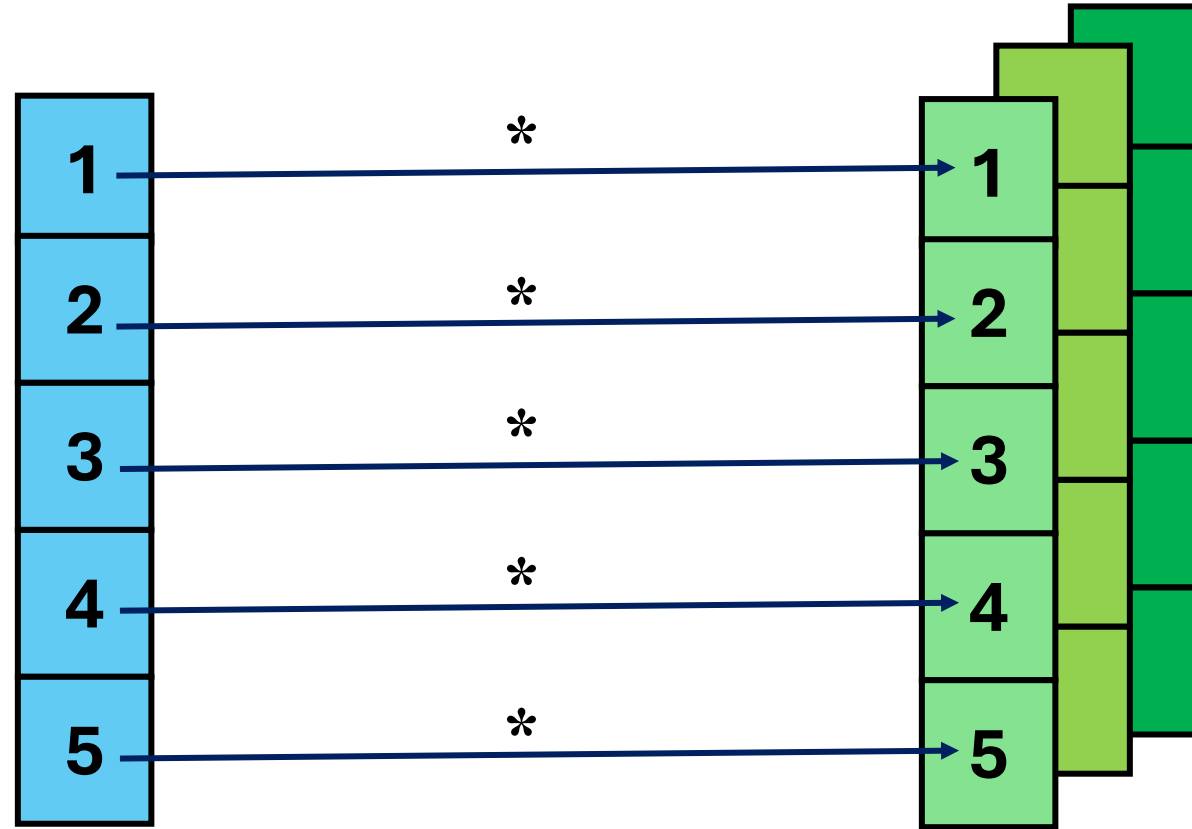
1
2
3
4
5

Leaky ReLU(x)

$x, x \geq 0$

$\alpha x, x < 0 (\alpha = 0.01)$

Fully Connect – Layer 2 :



Fully Connect – Layer 2

$$1*1 + 2*2 + 3*3 + 4*4 + 5*5 + \text{Bias2}$$



Softmax function :

For a vector $\mathbf{z} = [z_1, z_2, \dots, z_{n-1}, z_n]$, the Softmax function for the i-th element is defined as:

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}, \text{ where } \sigma(z)_i \text{ is the probability of the i-th element.}$$

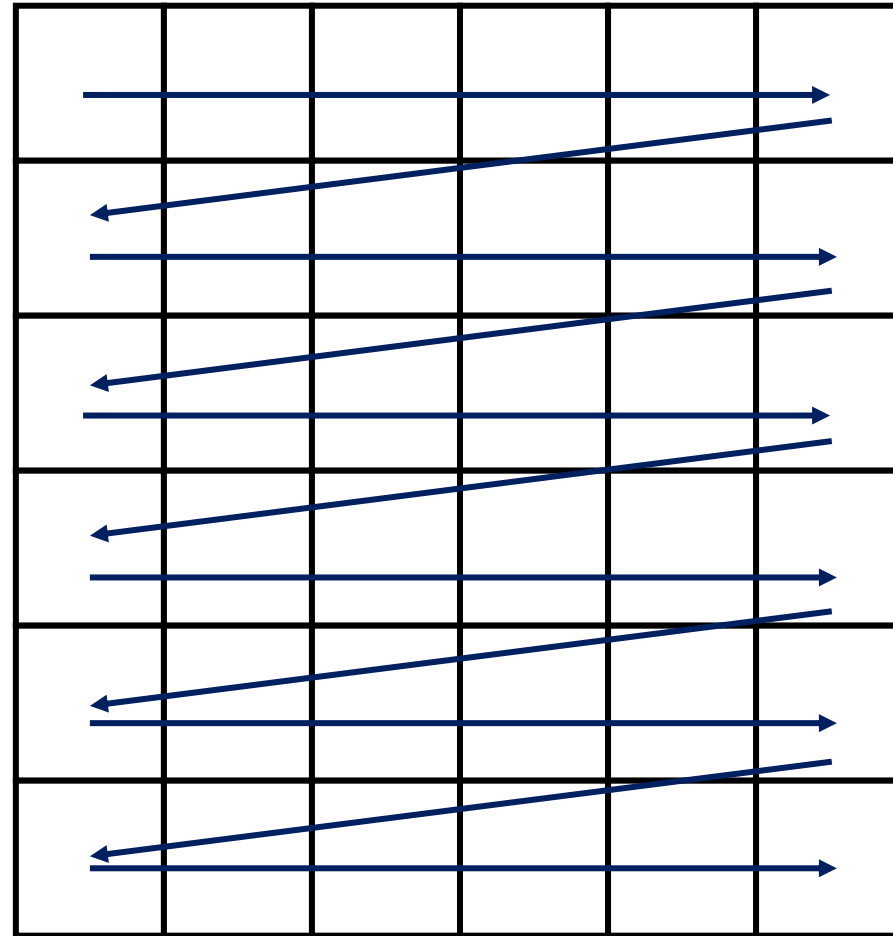
z_1	5.82
z_2	3.14
z_3	4.22

 $\longrightarrow \sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}} \longrightarrow$

σ_1	0.79
σ_2	0.05
σ_3	0.16

Task 1 – Cost-Constrained Kernel in CNN

Input Image :

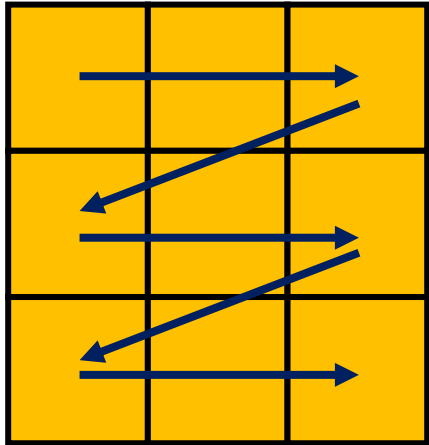


Image

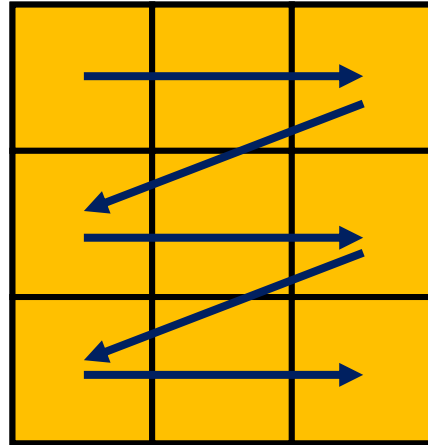
(6 x 6, 1 channel)

Input Kernel of A, B, C, D :

Kernel_ch1

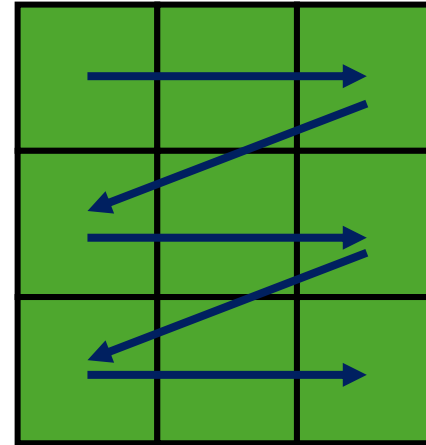


Kernel_A

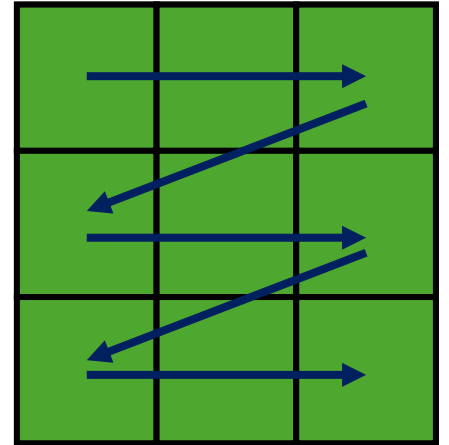


Kernel_B

Kernel_ch2



Kernel_C



Kernel_D

Input Capacity and Cost of A, B, C, D :

Capacity	Cost of A	Cost of B	Cost of C	Cost of D
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Replication Padding
(mode = 2'b00 or 2'b01) :

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36

1	1	2	3	4	5	6	6
1	1	2	3	4	5	6	6
7	7	8	9	10	11	12	12
13	13	14	15	16	17	18	18
19	19	20	21	22	23	24	24
25	25	26	27	28	29	30	30
31	31	32	33	34	35	36	36
31	31	32	33	34	35	36	36

Reflection Padding
(mode = 2'b10 or 2'b11) :

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36

8	7	8	9	10	11	12	11
2	1	2	3	4	5	6	5
8	7	8	9	10	11	12	11
14	13	14	15	16	17	18	17
20	19	20	21	22	23	24	23
26	25	26	27	28	29	30	29
32	31	32	33	34	35	36	35
26	25	26	27	28	29	30	29

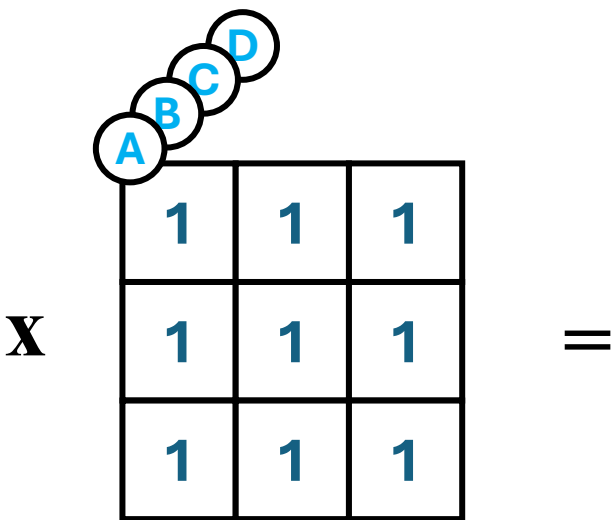
CNN :

1	1	2	3	4	5	6	6
1	1	2	3	4	5	6	6
7	7	8	9	10	11	12	12
13	13	14	15	16	17	18	18
19	19	20	21	22	23	24	24
25	25	26	27	28	29	30	30
31	31	32	33	34	35	36	36
31	31	32	33	34	35	36	36

8 x 8 Image (After Padding)

$1*1 + 1*1 + 2*1 + 1*1 + \dots + 8*1 = 30$

$1*1 + 2*1 + 3*1 + 1*1 + \dots + 9*1 = 36$

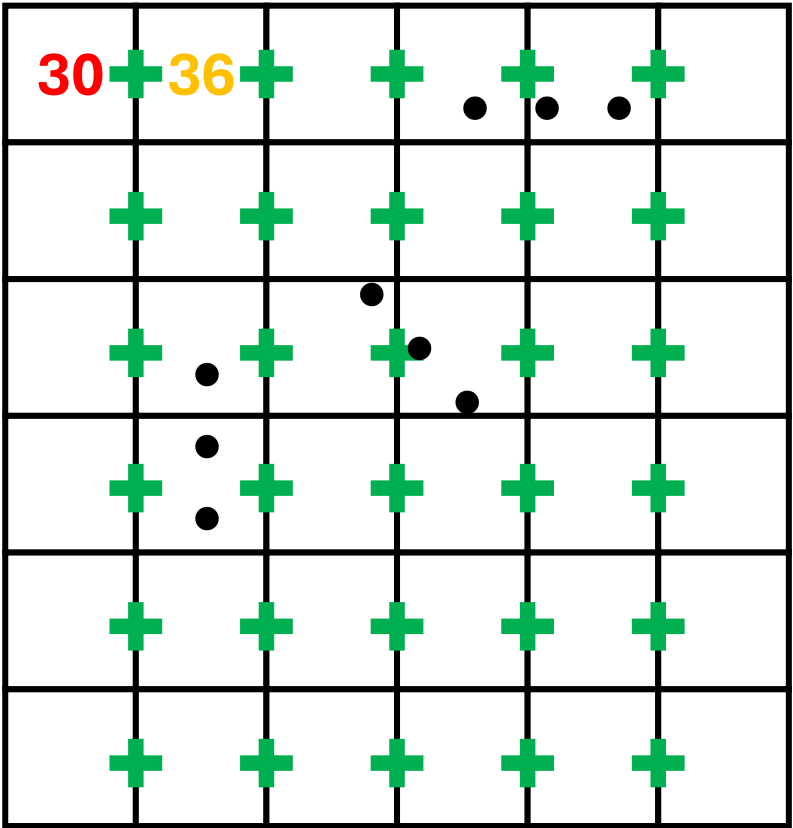


=

30	36				

6 x 6 out

Sum up all CNN results :



6 x 6 out

Select Kernel to achieve Max value :

The sum of the selected costs must be less than or equal to the available capacity

(sum of Cost $\leq$ Capacity)

For example :

capacity : **7**

cost of A : **5** , Afte kernel A CNN, sum of all CNN results = **2199**

cost of B : **2** , Afte kernel B CNN, sum of all CNN results = **3283**

cost of C : **1** , Afte kernel C CNN, sum of all CNN results = **2413**

cost of D : **1** , Afte kernel D CNN, sum of all CNN results = **2244**

We will select **Kernel B, C, D**

out = { **28'b0, 4'b0111** }

Selecting a kernel is **optional** — you may leave A, B, C, and D **all unselected**.